



Clinical Research

Machine Learning Reveals How Depression Influences Chest Pain Localisation and Its Predictive Value for Coronary Artery Disease

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ABSTRACT

Background: Depression is independently associated with chest pain and increased cardiovascular risk, regardless of coronary artery disease (CAD) status. However, limited research has examined how chest pain characteristics differ in individuals with and without depression. This study evaluated the relationship between depression and chest pain localisation to inform CAD risk assessment.

Methods: Data from the National Health and Nutrition Examination Survey from 2005 to 2020 were analysed for adults completing the chest pain questionnaire. Survey-weighted propensity score matching created depression-stratified cohorts. Chest pain localisation and inter-regional correlations were compared using survey design-adjusted methods. Random forest models with survey-weighted bootstrap replicates were used to estimate the relative importance of pain locations in predicting CAD, stratified by depression status.

Results: A total of 2208 individuals were matched (1104 per cohort). Depressed individuals more frequently reported chest pain in the lower sternum ($P = 0.045$), left chest ($P = 0.002$), and epigastrium

RÉSUMÉ

Contexte : Un lien indépendant a été établi entre la dépression d'une part et la douleur thoracique et un risque cardiovasculaire accru d'autre part, peu importe la présence ou non d'une coronaropathie. Les recherches ayant examiné comment les caractéristiques de la douleur thoracique diffèrent chez les personnes atteintes ou non de dépression sont néanmoins limitées. Dans cette étude, nous avons évalué le lien entre la dépression et le siège de la douleur thoracique afin d'affiner l'évaluation du risque de coronaropathie.

Méthodologie : Nous avons analysé les données recueillies entre 2005 et 2020 dans le cadre de l'enquête nationale sur la santé et la nutrition (National Health and Nutrition Examination Survey, NHANES, États-Unis) chez les adultes ayant rempli le questionnaire sur la douleur thoracique. Nous avons apparié les scores de propension après pondération en fonction des données de l'enquête pour obtenir des cohortes stratifiées selon la présence ou l'absence de dépression. Nous avons comparé le siège de la douleur thoracique et les corrélations interrégionales à l'aide de méthodes adaptées en fon-

Keywords: machine learning; depression; coronary artery disease; angina; chest pain

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Chest pain refers to a broad descriptor of discomfort and/or pressure in the sternum and surrounding areas, and though often associated with cardiac disease, can be secondary to a diverse range of pathologies. As such, clinicians and researchers have attempted to decode the characterisation of chest pain and the association with differing disease processes to aid in screening. Angina, or chest pain secondary to coronary artery disease (CAD), is a primary culprit of these symptoms, caused by an imbalance of oxygen supply and demand to the myocardium. Given the potentially lethal

($P = 0.039$). In depressed individuals, atypical pain regions (epigastrium, lower sternum, neck, arms) were more predictive of CAD, whereas in nondepressed individuals, typical regions (chest, upper sternum) were stronger predictors. These findings were robust to temporal validation and stricter definitions of depression.

Conclusions: Depression modifies the predictive value of chest pain localisation for CAD. A “depressed chest pain profile” involving nontraditional locations was more strongly associated with CAD in those with depression, and a more central “nondepressed pain profile” was more predictive in those without. These findings underscore the importance of integrating mental health context into chest pain evaluation.

consequences, this is of great concern for clinicians to identify in a timely manner.

A “typical” angina chest pain pattern has been defined for individuals with CAD, generally described as a retrosternal pressure-like pain, with radiation to the neck, jaw, and occasionally arms, and often associated with exertion. However, this pattern is not always reliable, and there is growing evidence that certain subpopulations, such as women and individuals with diabetes, may have a uniquely different experience of angina.¹⁻³

Depression has been identified to have an independent association with chest pain, even in individuals without obstructive CAD.⁴ Studies have shown depression severity to be significantly correlated with angina frequency after adjustment for CAD severity and antidepressant use, with improvement in depressive symptoms leading to improvement in angina.⁵ In addition, individuals with depressive symptoms are more likely to experience future cardiovascular events in CAD patients compared with those without.⁶

Growing recognition of the link between mental health and cardiovascular outcomes has led the American Heart Association to recommend routine depression screening in patients with CAD.⁷ However, there remains a gap in understanding how chest pain characteristics may differ in depressed individuals, with or without underlying CAD. A recent 2025 NHANES analysis by Yuan et al. explored the interaction between chest pain and depression risk in the context of coronary heart disease, but their focus was on predicting depression rather than characterising pain localisation.⁸ To date, no studies have specifically examined how depression may alter the diagnostic value of chest pain presentation for CAD.

In the present study, we aimed to use population-complex survey data collected in the National Health and Nutrition

Examination Survey (NHANES) database to investigate how depression influences chest pain localisation. We also aimed to assess the predictive value of chest pain regions for underlying CAD within depressed and nondepressed cohorts with the use of a novel machine learning approach to capture nonlinear relationships and account for the variability in the complex survey design. Our hope is that this population-level evidence can inform clinical decision making by tailoring chest pain assessment in patients with depression.

Résultats : Au total, 2 208 personnes ont été appariées (1 104 par cohorte). Les personnes dépressives ont signalé plus souvent des douleurs thoraciques dans le bas du sternum ($p = 0,045$), dans le côté gauche de la poitrine ($p = 0,002$) et dans la région épigastrique ($p = 0,039$). Chez les personnes dépressives, les sièges de douleur inhabituels (région épigastrique, bas du sternum, cou, bras) étaient plus prédictifs de la coronaropathie, tandis que chez les personnes non dépressives, les sièges habituels (thorax, haut du sternum) étaient des facteurs prédictifs plus puissants. Ces résultats se sont avérés fiables après validation temporelle et l'application de définitions plus strictes de la dépression.

Conclusions : La dépression modifie la valeur du siège de la douleur thoracique pour prédire la coronaropathie. Le profil de douleur thoracique caractérisé par des sièges inhabituels était plus fortement lié à la coronaropathie chez les personnes atteintes de dépression, tandis que le profil de douleur thoracique plus central était plus prédictif de coronaropathie chez les personnes non atteintes de dépression. Ces résultats soulignent l'importance de prendre en considération la santé mentale dans l'évaluation de la douleur thoracique.

Examination Survey (NHANES) database to investigate how depression influences chest pain localisation. We also aimed to assess the predictive value of chest pain regions for underlying CAD within depressed and nondepressed cohorts with the use of a novel machine learning approach to capture nonlinear relationships and account for the variability in the complex survey design. Our hope is that this population-level evidence can inform clinical decision making by tailoring chest pain assessment in patients with depression.

Methods

Study population and data preprocessing

Data were derived from NHANES, a continuous, cross-sectional survey of the noninstitutionalized US population. All NHANES data are publicly available and fully de-identified; therefore, this secondary analysis was deemed to be exempt from institutional ethics review.⁹ NHANES uses a complex multistage probability sampling design with clustering and stratification. We included adult participants (aged ≥ 40 years) from the 2005 to 2020 pre-Covid survey cycles who completed the cardiovascular questionnaire. Participants missing data for depression or CAD status were excluded. Covariates with missing data were retained and imputed, unless missingness exceeded 10%, in which case they were excluded from sensitivity analyses.

Missing data were evaluated (Supplemental Fig. S1). No participants had $> 10\%$ missing data overall. Three covariates exceeded 10% missingness and were excluded from sensitivity models. For all other variables, missingness was assumed to be at random and addressed using random forest imputation, a method suitable for nonlinear patterns of missingness.¹⁰

Key variable definitions

Depression was defined using a composite indicator: a Patient Health Questionnaire (PHQ-9) score ≥ 5 , current use of antidepressant medication, or a mental health visit within the past 12 months.¹¹ This approach was chosen to maximise sensitivity and capture a broad spectrum of individuals with any evidence of depressive symptoms or history of depression-related care. By including individuals with mild symptoms (PHQ-9 ≥ 5), we aimed to distinguish between any depression vs none, rather than limit to moderate or severe cases. This definition also allowed us to identify participants with treated or previously recognised depression who may not currently report high symptom burden. However, to ensure robustness, we completed a secondary sensitivity analysis using a stricter depression definition (PHQ-9 ≥ 10). CAD was defined by self-reported history of any of the following: diagnosis of CAD, angina, or myocardial infarction. Self-reported medical history—particularly for CAD—has an estimated sensitivity of 84% and specificity of 96%, indicating a relatively low risk of misclassification.^{12,13}

Demographic and health covariates (eg, age, sex, body mass index, smoking, income) were included for adjustment and are detailed in [Supplemental Appendix S1](#). NHANES design variables (stratum, cluster, weights) were modified per Centers for Disease Control and Prevention guidelines to account for multiple cycles.

Chest pain assessment

Participants reporting chest pain were asked to localise it using a diagram with multiple selectable regions: right arm, right chest, neck, upper sternum, lower sternum, left chest, left arm, and epigastrium. They were also asked about severe chest pain (> 30 min) and dyspnoea with exertion.

Propensity score matching

To control for confounding, we applied survey-weighted 1:1 nearest-neighbour propensity score matching without replacement, using 18 covariates ([Supplemental Appendix S2](#)).¹⁴ Matching was performed within strata; strata with a single primary sampling unit were excluded. Propensity scores were estimated via survey-weighted logistic regression. Weights were transferred from treated to control units.

Covariate balance was assessed by means of population-level standardised mean differences, with $\leq 10\%$ considered adequate. Additional comparisons using survey-weighted medians, interquartile ranges (IQRs), and proportions are reported in [Table 1](#) and [Supplemental Figure S2](#).

A full-cohort sensitivity analysis (without matching) was conducted to assess robustness of results.

Sample size and power estimation

Effective sample size (ESS) was estimated using the maximum design effect (maxDEFF), derived from key variables (CAD, depression, pain regions).¹⁵ *Post hoc* ESS was calculated as $ESS = \text{nominal sample size} / \text{maxDEFF}$. The minimum required sample size to detect a small effect (Cohen $d = 0.3$) at 80% power and $\alpha = 0.05$ was ~ 175 .

Statistical analyses

Proportions of chest pain localisation were compared between cohorts by means of the Rao-Thomas adjusted Wald test.¹⁶ Total numbers of pain regions per participant were compared by means of survey-weighted t tests.

Predictive modelling of CAD

We modelled the probability of CAD using a bootstrapped random forest approach tailored to complex survey data. Models included all covariates used in matching to adjust for confounding and enhance interpretability. A workflow of this bootstrap survey-weighted random forest approach is summarised in [Figure 1](#).

Survey-design structure was preserved by using replicated weights generated by the survey package.¹⁷ Grid search with cross-validation was used to optimise hyperparameters, including the number of features randomly selected at each split (mtry) and the minimum terminal node size (min.node.size). For each of 250 bootstrapped datasets, we trained a random forest (100 trees), applied 5-fold cross-validation, and computed 0.632 bootstrapped performance metrics (area under the receiver operating characteristic curve [AUC], G-mean, Brier score). We conducted temporal model validation by training the model on participants from 2005–2016 and evaluating its performance on an independent test cohort from 2016–2020. ALE (accumulated local effect) plots were used to interpret predictor effects.¹⁸

We computed ALE for each binary pain region and quantified its average effect (Δ ALE) on CAD probability. Distributions of Δ ALE were generated from bootstrapped models, and 95% confidence intervals were used to identify significant regional predictors.

We defined cohort-specific chest pain profiles by aggregating Δ ALEs across pain regions, resulting in the Depressed Chest Pain Profile (DCPP) and Nondepressed Chest Pain Profile (NDCPP). Within- and between-cohort comparisons of these profiles were conducted using bootstrapped confidence intervals.

We performed decision curve analysis (DCA) within the depressed cohort by comparing 2 predictive models: one incorporating all chest pain regions, and another excluding regions characteristic of depression-associated pain. This approach assessed whether the inclusion of depression-specific pain regions provided additional clinical utility in identifying patients at risk for CAD.

Simulation

To validate this method, we simulated a complex survey dataset with nested strata and known effects. We compared our survey-weighted bootstrapped forest model against an unweighted model. Recovery of true effect sizes was assessed across bootstrap samples, and visualised results are shown in [Supplemental Figure S3](#).

Software and code

All analyses were conducted in R (v4.4.3).¹⁹ Packages included survey, caret, ranger, missForest, and iml.^{20–24} Code and additional resources are provided in [Supplemental Appendix S3](#). In accordance with best practices for

Table 1. Patient demographic and health variables for depression-stratified cohorts before and after survey-weighted propensity score matching

		Survey cohorts (n = 6790)			Matched cohorts (n = 2208)		
		Depressed (n = 1118)	Nondepressed (n = 5672)	SMD	Depressed (n = 1104)	Nondepressed (n = 1104)	SMD
Age, y	40-49	28.2	27.9	0.006	28.3	27.1	0.028
	50-59	36.6	27.9	0.186	36.4	35.4	0.021
	60-69	20.6	21.9	-0.030	20.7	21.6	-0.022
	70-79	10.5	14.4	-0.119	10.5	11.8	-0.043
	≥ 80	4.1	7.9	-0.160	4.1	4.1	0.002
Sex	Female	32.2	52.9	-0.428	31.9	36.6	-0.100
	Male	67.8	47.1	0.428	68.1	63.4	0.100
Race/ethnicity	Mexican American	3.8	6.3	-0.113	3.8	3.5	0.019
	Other Hispanic	5.7	4.8	0.039	5.7	5.5	0.012
	Non-Hispanic White	75.1	69.7	0.121	75.0	74.5	0.011
	Non-Hispanic Black	9.1	12.3	-0.103	9.1	9.9	-0.029
	Other	6.3	6.9	-0.026	6.4	6.6	-0.010
Country of birth	USA	91.8	85.7	0.196	91.8	91.3	0.015
	Outside USA	8.2	14.3	-0.196	8.2	8.7	-0.015
Education level	< 9th grade	5.8	7.7	-0.079	5.8	5.1	0.028
	9th-11th grade (no diploma)	11.0	11.0	0.001	11.2	11.6	-0.013
	High school grad/GED	28.9	25.3	0.081	28.7	28.7	0.000
	Some college or AA	36.4	30.0	0.136	36.6	35.0	0.034
	College grad	18.0	26.0	-0.195	17.8	19.7	-0.047
Income (as proportion of poverty threshold)	0-0.99	23.7	14.3	0.242	23.8	21.8	0.048
	1-1.99	27.1	20.5	0.153	27.1	26.6	0.012
	> 2	49.2	65.2	-0.326	49.1	51.6	-0.051
Health insurance coverage		91.1	86.7	0.140	91.1	90.4	0.023
BMI, kg/m ²		30.9 (26.2-35.7)	28.9 (25.2-33.3)	0.252	31.1 (26.2-35.7)	30.0 (26.1-35.7)	0.082
Moderate to vigorous physical activity at least once a week		66.1	53.0	0.269	66.0	62.5	0.073
Hypertension		60.2	50.6	0.194	60.2	56.2	0.081
High cholesterol		63.2	53.4	0.199	63.6	61.8	0.038
Diabetes	No	76.7	82.4	-0.140	76.5	78.8	-0.055
	Borderline	3.5	3.2	0.016	3.5	3.0	0.029
	Diet-controlled	2.6	2.5	0.006	2.6	2.3	0.019
	Medication-controlled	9.4	7.3	0.074	9.5	9.1	0.016
	Insulin ± medications	7.8	4.6	0.132	7.9	6.8	0.040
Use of any illicit drug in the past		56.1	41.0	0.306	55.9	53.8	0.042
Alcohol-use (avg. no. drinks/day)		2.0 (1.0-3.0)	2.0 (1.0-3.0)	0.009	2.0 (1.0-3.0)	2.0 (1.3-3.0)	0.043
Current smoker	Every day	38.4	25.0	0.291	38.0	34.5	0.073
	Some days	5.1	3.0	0.106	5.2	5.0	0.009
	No	56.5	72.0	-0.328	56.8	60.5	-0.075
Smoked at least 100 cigarettes in lifetime		37.1	46.2	-0.184	37.5	39.6	-0.044
NHANES cycle	2005-2006	10.2	14.7	-0.136	9.5	9.2	0.012
	2007-2008	10.9	13.4	-0.077	10.6	12.2	-0.050
	2009-2010	8.8	11.0	-0.076	8.8	8.2	0.022
	2011-2012	8.2	11.5	-0.111	8.3	9.4	-0.038
	2013-2014	15.4	11.2	0.125	15.6	15.3	0.008
	2015-2016	15.0	14.4	0.018	15.2	16.1	-0.026
	2017-2020 (pre-Covid)	31.6	23.9	0.172	32.0	29.6	0.051

Values represent survey-weighted proportion, ie, %, or median (interquartile range).

AA, Associate of Arts; BMI, body mass index; GED, General Educational Development; NHANES, National Health and Nutrition Examination Survey; SMD, standardized mean difference.

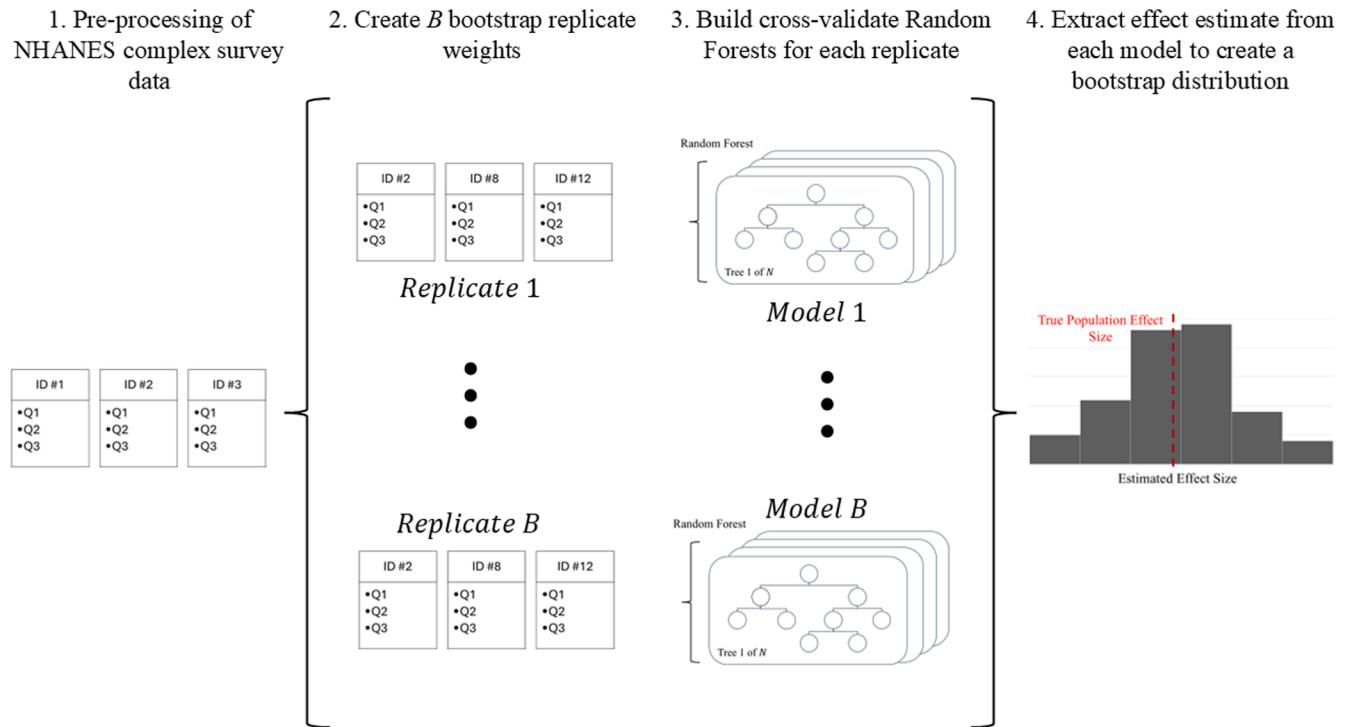


Figure 1. Bootstrap survey-weighted random forest workflow.

transparency and reproducibility in AI-based prediction model research, we followed the TRIPOD-AI reporting guideline; the completed checklist is included in the appendix²⁵ (Supplemental Table S1).

Results

In total, 2208 survey respondents (approximate representative US population size: $\sim 11,967,239$) were included in the matched analysis, with 1104 in each depression-stratified cohort (Fig. 2). Calculated maxDEFF was 2.29, resulting in a minimum ESS of 482 per cohort, larger than the required sample size needed for an adequately powered effect estimate. Baseline demographics and clinical characteristics before and after matching and between depressed and nondepressed groups are summarised in Table 1. Satisfactory balance was achieved after matching, with a standardised mean difference $\leq 10\%$ for all covariates (Table 1, Supplemental Fig. S2).

Figure 3 displays survey-weighted box-and-whisker plots comparing the proportion of individuals localising chest pain to each anatomic region between the depressed and nondepressed cohorts. The proportions of individuals localising chest pain to each region were compared between cohorts, with results demonstrating a significantly larger proportion of depressed individuals localising pain to the lower sternum ($P = 0.045$), left chest ($P = 0.002$) and epigastrium ($P = 0.039$) compared with nondepressed respondents. There was no significant difference in proportions among other pain regions (Fig. 3). On average, depressed individuals localised pain to more regions than

nondepressed individuals (1.70 vs 1.58), although this difference was insignificant ($P = 0.297$).

A final ensemble model was built to predict CAD for each depression cohort with both demonstrating good discrimination and calibration (depressed model: AUC 0.813, G-mean 0.781, Brier 0.131; non-depressed model: AUC 0.828, G-Mean 0.759, Brier 0.129). Temporal validation demonstrated strong model performance, with AUC 0.842, G-mean 0.809, and Brier 0.124. Results from the ALE analysis demonstrated that in depressed individuals, pain in the right arm (4.6% vs -0.0003% , Δ 4.6% [IQR 1.0%-9.8%]), left arm (6.3% vs 3.0%, Δ 3.3% [IQR -2.5% to 9.1%]), neck (6.6% vs 2.9%, Δ 3.7% [IQR -2.5% to 11.6%]), lower sternum (7.0% vs 2.3%, Δ 4.7% [IQR 0.4%-9.9%]), and epigastrium (1.9% vs 0.9%, Δ 1.1% [IQR -1.7% to 4.4%]) were on average more powerful for predicting CAD compared with nondepressed individuals, with right arm pain and lower sternal pain showing significant difference. In contrast, nondepressed individuals localising pain to the upper sternum (2.2% vs 4.7%, Δ -2.5% [IQR -5.9% to 0.6%]), right chest (1.6% vs 12.4%, Δ -10.8% [IQR -20.8% to -2.4%]), and left chest (2.0% vs 5.2%, Δ -3.2% [IQR -7.7% to 0.2%]) were on average more predictive of CAD compared with depressed individuals, with right chest pain showing significant difference (Fig. 4). These results are summarised in Figure 5A using heat maps to represent the average marginal effect (Δ ALE) of each pain region on CAD prediction in each cohort, with colour intensity indicating the strength and direction of association within each group. In addition, severe pain lasting > 30 minutes was on average stronger at predicting

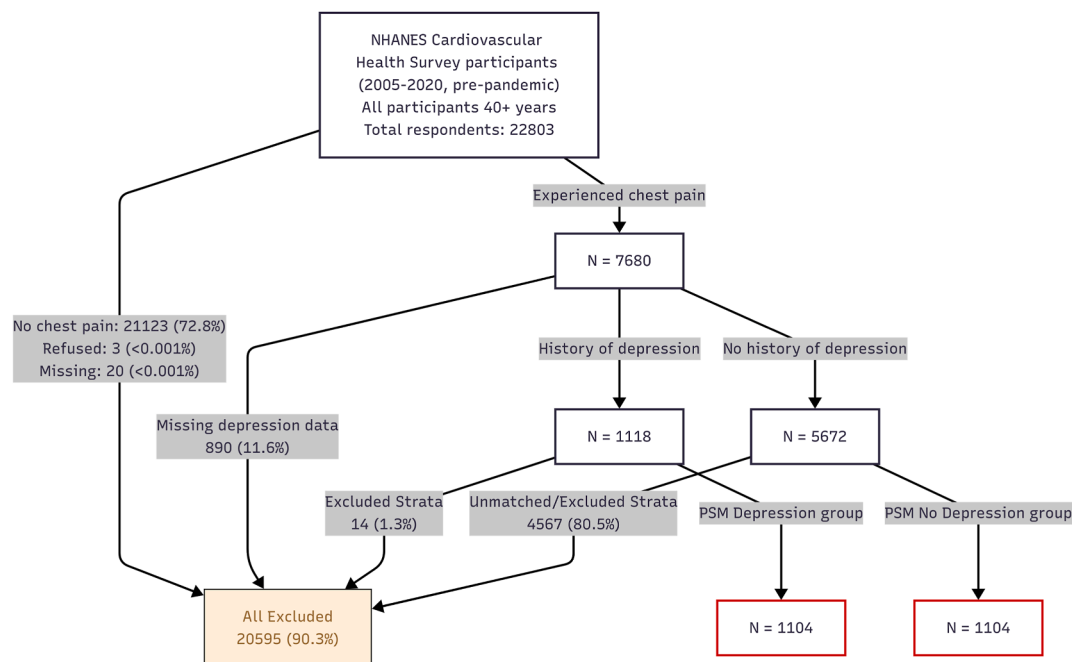


Figure 2. Flowchart of data subsetting of NHANES Cardiovascular Health Survey participants and final cohort nominal sample size after propensity score matching. NHANES, National Health and Nutrition Examination Survey; PSM, propensity score matched.

CAD in nondepressed individuals compared with depressed individuals (9.7% vs 14.5%, $\Delta -4.8\%$ [IQR -11.5% to 1.6%]), whereas the opposite is seen for shortness of breath on stairs or incline (2.2% vs -0.0001% , $\Delta 2.2\%$ [IQR -0.2% to 5.3%]), although neither was significantly different between cohorts.

DCPP was identified as pain in the left arm, right arm, neck, lower sternum, and epigastrium, and NDCPP was

defined as pain in the left chest, right chest, and upper sternum. Within-cohort differences were significantly different between cohort-specific pain profiles for both cohorts (depressed: $\Delta 3.2\%$ [0.8%-5.8%]; nondepressed: $\Delta -5.6\%$ [-6.9% to -2.3%]). Across-cohort differences were also significantly different between cohort-specific pain profiles (DCPP: $\Delta 3.4\%$ [1.2%-6.0%]; NDCPP: $\Delta -5.6\%$ [-6.7% to -2.0%]) (Fig. 5B).

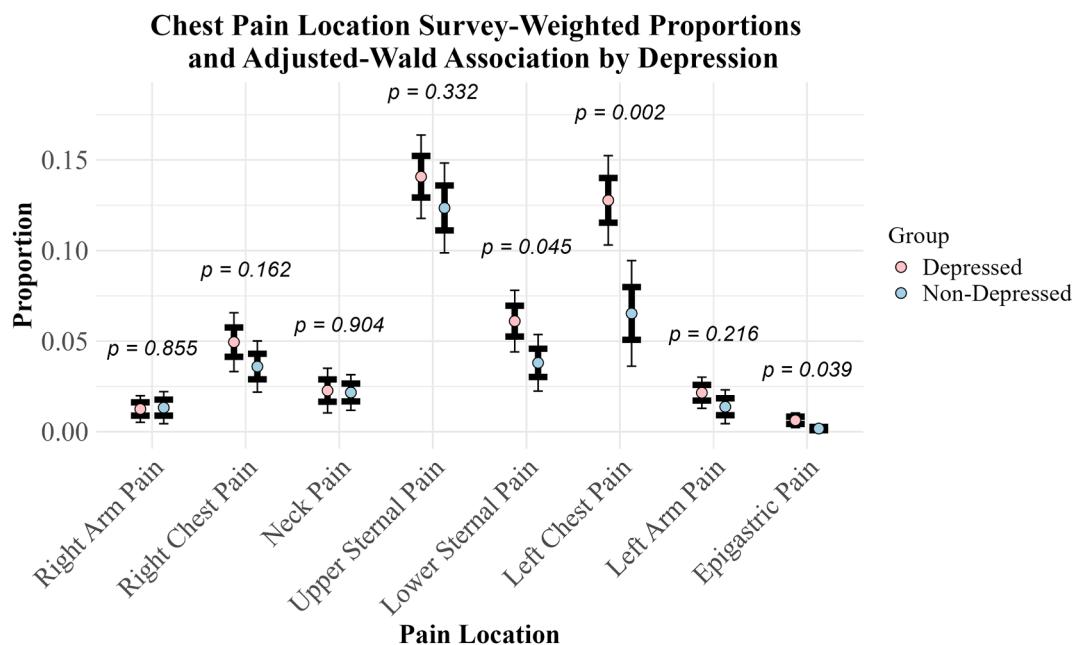


Figure 3. Chest pain location survey-weighted proportions and adjusted Wald association by depression.

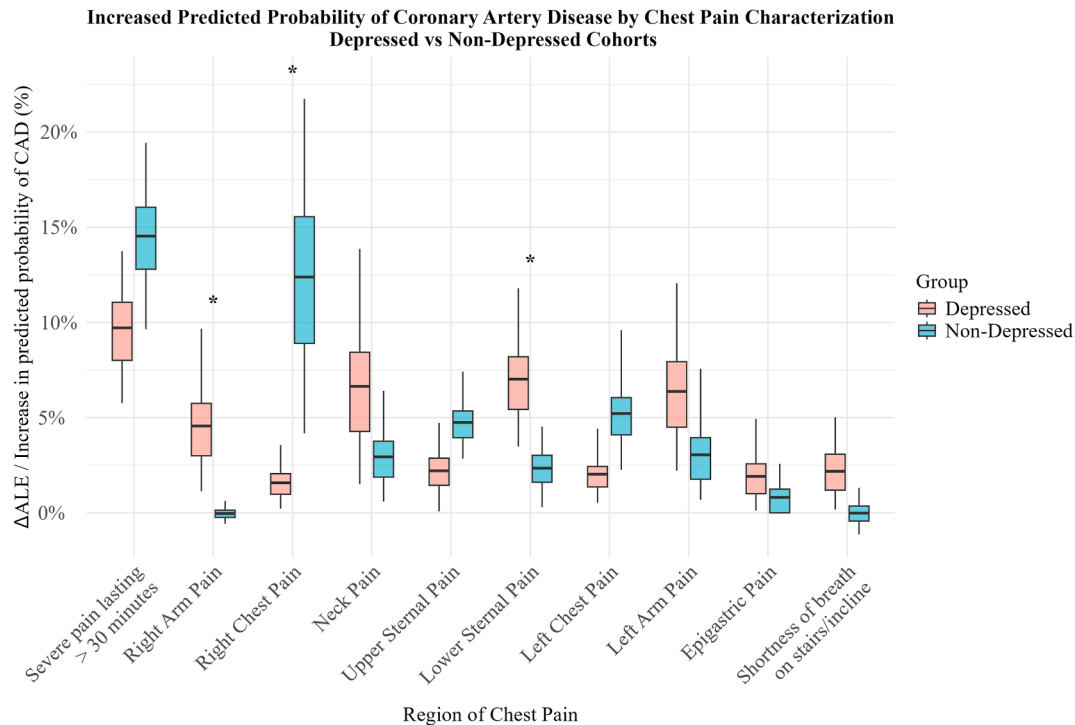


Figure 4. Increased predicted probability of coronary artery disease (CAD) by chest pain characterization in depressed vs nondepressed cohorts. *Significant difference between depression-stratified cohorts. ALE, accumulated local effects.

Using a stricter definition of depression ($\text{PHQ-9} \geq 10$), similar results emerged, with there being significant differences within (depressed: Δ 1.5% [IQR 0.8%–3.8%]; nondepressed: Δ –2.7% [IQR –3.3% to –2.0%]) and across (DCPP: Δ 1.9% [IQR 1.2%–2.5%]; NDCPP: Δ –2.2% [IQR –2.9% to –1.6%]) depression-stratified cohorts in pain profiles and their predictive value of CAD. Of note, however, neck pain was more strongly associated with CAD in nondepressed than in depressed individuals, which is contrary to the findings in the original analysis (Supplemental Fig. S4).

Missing-data sensitivity analysis was performed excluding the 3 variables with $> 10\%$ missingness (alcohol use, smoking status, illicit drug use), with results similar to the primary analysis, demonstrating similarly significant differences in chest pain regions between depression-stratified cohorts (Supplemental Fig. S5). Full-cohort sensitivity analysis also demonstrated the same Δ ALE cohort-specific pain profiles, validating the matched primary analysis results (Supplemental Fig. S6).

On DCA, the model including all chest pain regions demonstrated net benefit for predicting CAD in patients with chest pain and depression. However, there was no significant difference in net benefit between the full model and the model excluding depression-specific chest pain regions, suggesting limited added value of those regions in clinical decision making (Supplemental Fig. S7).

Discussion

Our findings provide novel insight into how depression modifies chest pain localisation and its diagnostic

implications for CAD. Although previous population-based studies have established a broad association between depression and chest pain, the present study adds a crucial dimension: Among individuals reporting chest pain, depressed individuals reporting pain in “atypical” regions—specifically the neck, epigastrium, arms, and lower sternum—had a stronger association with CAD than nondepressed individuals did. Conversely, more “typical” angina locations, such as the chest and upper sternum, were stronger predictors of CAD among nondepressed individuals. Importantly, these differences in pain patterns were statistically significant both within and across cohorts, reinforcing the notion that depression not only alters pain perception but also reshapes its diagnostic significance.

These findings may have practical clinical relevance. In emergency or outpatient settings, chest pain localised to noncardiac regions such as the epigastrium are often considered to be lower risk. However, our results suggest that in depressed individuals, these same pain patterns may signal a higher underlying risk of CAD. This has important implications for diagnostic triage and risk stratification, where symptom location is a central component of decision making. Clinical pathways—such as those used in emergency department chest pain evaluations or stress testing protocols—may benefit from incorporating mental health status to more accurately interpret pain localisation. For example, risk calculators or electronic triage systems could be refined to weigh atypical pain patterns more heavily in patients with depressive symptoms, potentially expediting necessary cardiac investigations in a subgroup that might otherwise be underassessed.

The potential biological mechanisms underlying these depression-specific pain localisations are multifactorial.

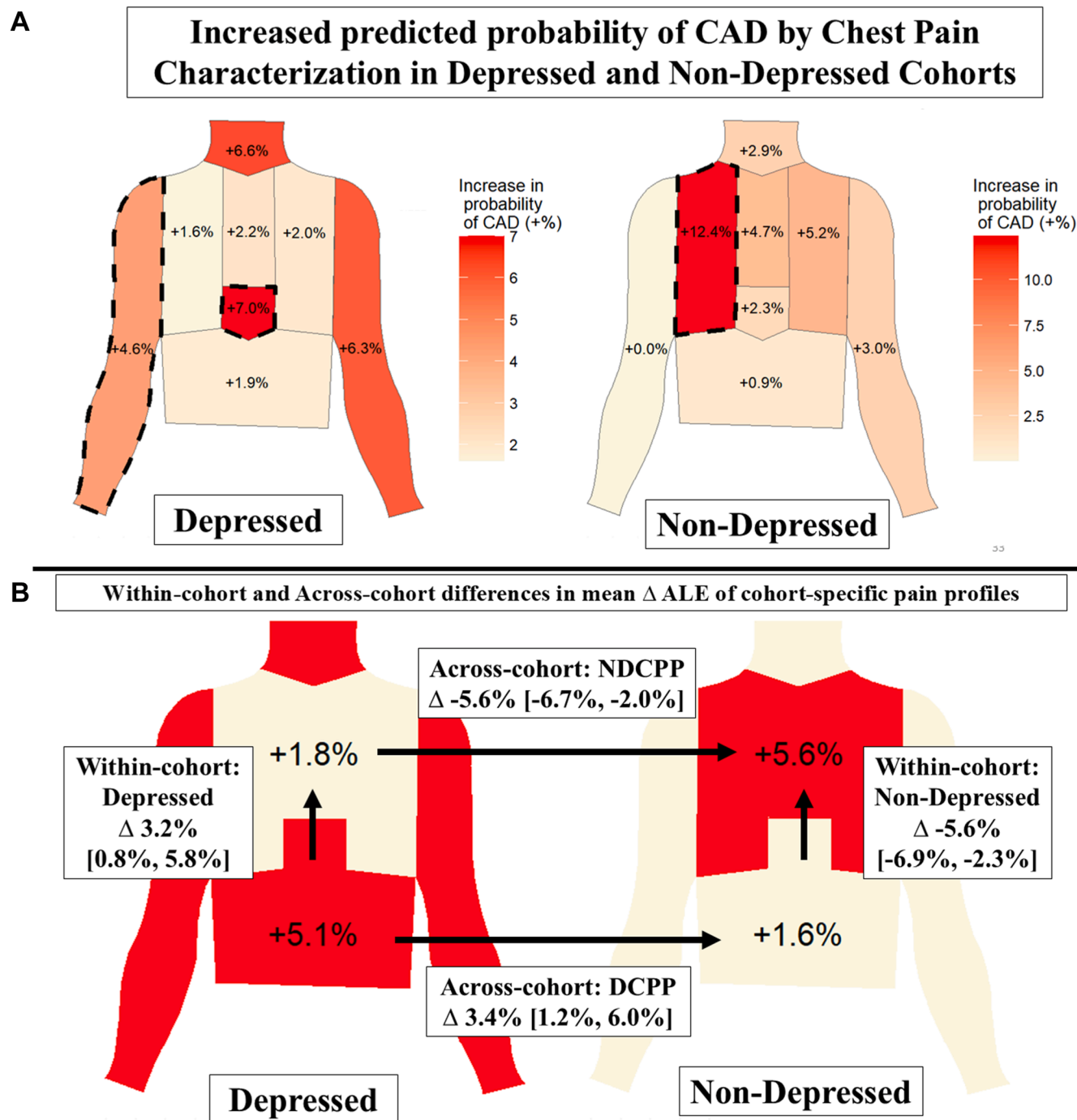


Figure 5. (A) Increased predicted probability of coronary artery disease by chest pain characterization in depressed vs nondepressed cohorts: heatmap. Significant differences between depression-stratified cohorts are outlined in dashed lines. **(B)** Within-cohort and across-cohort differences in mean Δ ALE of cohort-specific pain profiles. ALE, accumulated local effects; CAD, coronary artery disease; DCPD, depressed cohort pain profile; NDCPP, nondepressed cohort pain profile.

Depression is associated with dysregulation of neurotransmitters like norepinephrine and serotonin, altered hypothalamic-pituitary-adrenal axis activity, and changes in pain-processing networks in the brain, such as the anterior cingulate cortex and insula. These changes not only lower the threshold for pain perception but also distort the spatial representation of somatic symptoms.²⁶⁻²⁹ Central

sensitisation—an amplification of pain signalling within the central nervous system—may further contribute to this phenomenon by enhancing the perception of diffuse or atypical pain and diminishing the salience of classic ischemic regions such as the upper sternum.³⁰ In addition, depression is associated with attentional bias toward negative internal states, which may reduce awareness or accurate localisation of

pain in prototypical regions.³¹ Moreover, a growing body of evidence links depression with systemic inflammation, which further disrupts neurotransmitter metabolism and neural signalling. Elevated inflammatory markers (eg, CRP, IL-6) are associated with increased sensitivity to visceral pain and altered interoception. This may explain why visceral pain regions, such as the epigastrium, or regions associated with referred pain pathways, such as the neck and arms, are more commonly reported by depressed individuals and show stronger associations with CAD in the present study.³²

Methodologically, our use of survey-weighted propensity score matching and bootstrapped random forest modelling provides a robust population-representative framework for analysing nuanced patterns within the NHANES dataset. The use of ALE allowed us to quantify the marginal contribution of each pain region to CAD prediction, stratified by depression status. This not only strengthened model interpretability but revealed distinct cohort-specific pain profiles that would be obscured by traditional regression models.

Compared with the existing literature, our results are consistent with previous findings that depression increases angina frequency and lowers chest pain thresholds. However, this study is among the first to demonstrate that the topography of chest pain—not just its presence or severity—is meaningfully altered by depression and clinically relevant for CAD prediction. By moving beyond global pain metrics to examine regional patterns, we provide actionable insights for tailoring diagnostic approaches.

Despite its strengths, this study has several limitations. The cross-sectional design of NHANES precludes causal inference, and CAD status was self-reported, which may lead to under- or overestimation of disease burden. Although we adjusted for key confounders and accounted for the complex

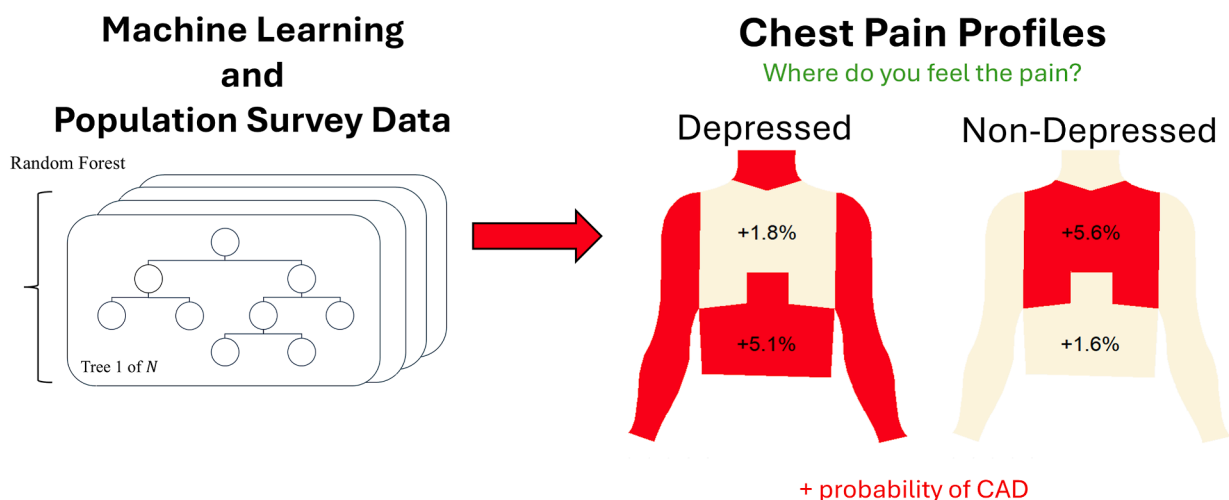
survey design and nonlinear relationships with the use of machine learning, residual confounding and challenges in interpreting black-box models may persist.

Generalisability is another consideration. NHANES represents a broad community-dwelling US adult population and includes individuals reporting chest pain regardless of whether they sought care, enhancing relevance to real-world symptom prevalence. By restricting our cohort to adults ≥ 40 years old with chest pain, we enrolled a predominantly male population, resulting in an inverse sex ratio of depression prevalence compared with the general population. This may affect the generalisability of our findings at a population level.

In addition, the findings may not directly translate to clinical settings such as emergency departments, where patient characteristics and acuity differ. This is evident from our decision curve analysis: Although specific chest pain regions exhibit differential predictive value for CAD in depressed vs nondepressed individuals, these differences may not yield clinically meaningful improvements in decision making. This suggests that although stratification by depression status refines our understanding of symptom profiles, its utility for guiding individual-level clinical decisions may be limited, and further validation in external datasets is warranted.

Although the results likely generalise to middle-aged and older adults in other high-income countries, caution is warranted when extrapolating to younger populations or different health care systems. In addition, differences in health care-seeking behaviour among individuals with depression, as well as the potential cardiometabolic effects of antidepressant use, were not accounted for in this analysis and may represent additional sources of confounding. The counterintuitive observation that right-side chest pain was

Depression may alter experience of CAD-specific chest pain



Central Illustration. Depression may alter experience of coronary artery disease (CAD)—specific chest pain. In nondepressed individuals, typical central chest regions are more predictive, whereas in depressed individuals, atypical regions such as the arms, neck, lower sternum, and epigastrium carry greater diagnostic weight. These cohort-specific “pain profiles” were identified with the use of survey-weighted machine learning models, underscoring the importance of integrating mental health into CAD risk assessment.

more predictive than left-side pain in nondepressed patients highlights these limitations, because residual confounding, referral or diagnostic bias, and reliance on self-reported symptoms could all contribute to such unexpected associations. Prospective studies are needed to evaluate whether incorporating depression status into chest pain risk assessment tools could improve the diagnostic yield for CAD, particularly in settings of acute clinical evaluation.

Conclusion

This study reinforces the established link between depression and chest pain while revealing how depression shifts the spatial pattern of chest pain localisation in ways that may be clinically meaningful (Central Illustration). In depressed individuals, atypical pain regions carry greater diagnostic weight for CAD, suggesting the need to recalibrate how clinicians interpret chest pain in this population. These findings support a more personalised approach to CAD risk assessment—one that integrates mental health into the evaluation of somatic symptoms and ultimately enhances diagnostic precision for a vulnerable and often overlooked subgroup.

Ethics Statement

The study used publicly available, de-identified data from the National Health and Nutrition Examination Survey (NHANES). NHANES protocols are approved by the National Center for Health Statistics Research Ethics Review Board, and every participant provided written informed consent. Because this analysis used de-identified, publicly available data, additional institutional ethics approval was not required.

Patient Consent

The authors confirm that patient consent is not applicable in this submission. All data were obtained from the strictly de-identified public NHANES database.

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Supplementary Material

To access the supplementary material accompanying this article, visit the online version of the *Canadian Journal of Cardiology* at www.onlinecjc.ca and at <https://doi.org/10.1016/j.cjca.2025.09.017>.